

Kubernetes Deployment Assignment

Batch 03 - Assignment 3 - Kubernetes-Ramana Chandrakiran Reddy

Introduction

This assignment demonstrates deployment and management of an application using Kubernetes. The project includes containerization using Docker, deployment using Kubernetes resources, rolling updates, and health monitoring using probes.

Objective

The main objective of this assignment is to understand:

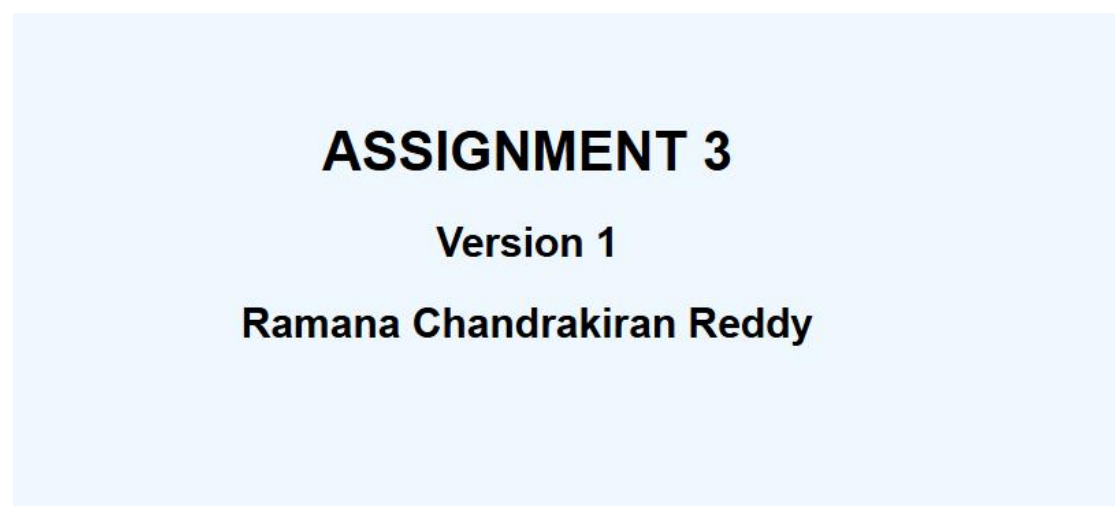
- Pod creation
- ReplicaSet management
- Deployment management
- Rolling updates
- Kubernetes Services
- Container health checks using probes

Frontend Application

The application is a simple frontend UI created using HTML and CSS.

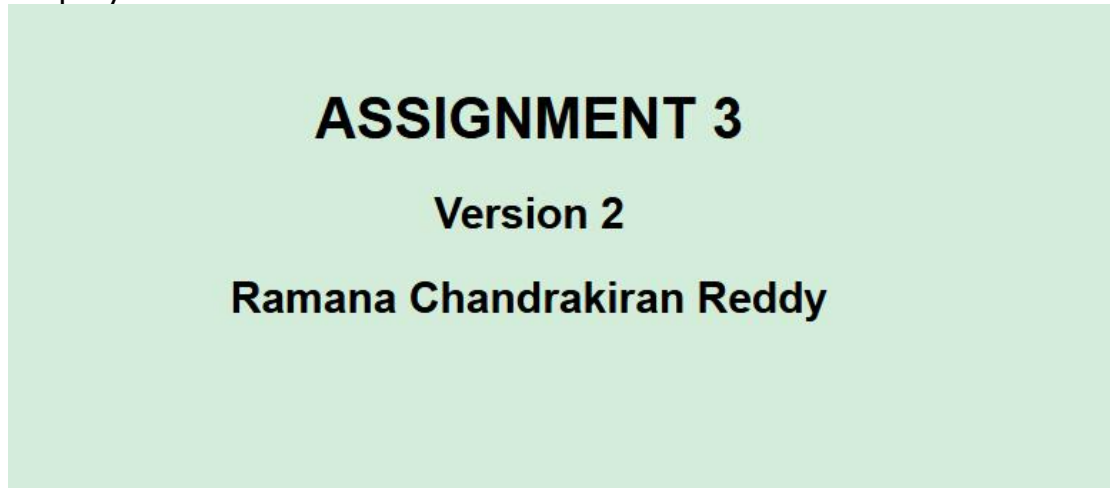
Version 1

Displays:



Version 2

Displays:



Docker Setup

Dockerfile

```
FROM nginx:alpine
COPY . /usr/share/nginx/html
EXPOSE 80
```

Docker Build Commands

Build v1

`docker build -t ramanachandrakiranreddy/k8s-webapp:v1 .`

```
PS C:\Users\npc\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\V1> docker build -t ramanachandrakiranreddy/k8s-webapp:v1 .
[+] Building 6.8s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 115B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 399B
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:5616878291a2eed594aee8db4d4de5878cf7edcb475e59193904b198d9b830de
=> resolve docker.io/library/nginx:alpine@sha256:5616878291a2eed594aee8db4d4de5878cf7edcb475e59193904b198d9b830de
=> sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f 1.40kB / 1.40kB
=> sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d 20.25MB / 20.25MB
=> sha256:453da7db72e7738c4e201ec3f3c4a5f7751adb5a6f127792c1cd001a780721 1.21kB / 1.21kB
=> sha256:701ff50d2644f7471c0d5a4ffa3447b0b5b7bbfd4b203c6fb1dc6a637722ae24c89241 404B / 404B
=> sha256:aee4e5b3865ee4d98545b6b49f9e8ab3b9b6ed114a2eafdc7a22e984ea5a68 956B / 956B
=> sha256:583599bb7d382e9e986a9ff65204981307bfa71670a29f4190eedae93c4858be 627B / 627B
=> sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91afff80deace02b6f2f26 1.87MB / 1.87MB
=> sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff98763195a704a5adda577bb 3.86MB / 3.86MB
=> extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff98763195a704a5adda577bb
=> extracting sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91afff80deace02b6f2f26
=> extracting sha256:583599bb7d382e9e986a9ff65204981307bfa71670a29f4190eedae93c4858be
=> extracting sha256:aee4e5b3865ee4d98545b6b49f9e8ab3b9b6ed114a2eafdc7a22e984ea5a68
=> extracting sha256:701ff50d2644f7471c0d5a4ffa3447b0b5b7bbfd4b203c6fb1dc6a637722ae24c89241
=> extracting sha256:453da7db73e7738c4e201ec3f3c4a5f7751adb5a6f127792c1cd001a780721
=> extracting sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f
=> extracting sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
=> exporting to image
=> exporting layers
=> exporting manifest sha256:7f1ead4b1c6f63bd7925e9759725fee9f311b0c388e8d94088c121b1f8797703
=> exporting config sha256:da4540215ec24ba330a885eb29631fffd8b66f41d7e0a91523904a0796a4a92c
=> exporting attestation manifest sha256:e978b9d7dc0c71c008ffae0f5edf58f8f9c10f1d629fb1b9e4d0b672cd0b1f8c
=> exporting manifest list sha256:a8d94fe468a3805e88cb20c24e10b299c471da5cd5152f4b480d40a9884eda38
=> naming to docker.io/ramanachandrakiranreddy/k8s-webapp:v1
=> unpacking to docker.io/ramanachandrakiranreddy/k8s-webapp:v1
```

Build v2

docker build -t ramanachandrakiranreddy/k8s-webapp:v2 .

```
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\V2> docker build -t ramanachandrakiranreddy/k8s-webapp:v2 .
[+] Building 1.3s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 115B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> => transferring context: 28
=> [internal] load build context
=> => transferring context: 310B
=> CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:5616878291a2eed594aee8db4d4ade5878cf7edcb475e59193904b198d9b830de
=> => resolve docker.io/library/nginx:alpine@sha256:5616878291a2eed594aee8db4d4ade5878cf7edcb475e59193904b198d9b830de
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:ad85df1a3c7340810e86c0e3a44e85187bb2960ed278cc1a4d980cf35add1edf
=> => exporting config sha256:ecf8fd3ac13a66b466a391c50c21641e225ba0049f081ddbc5cf24d06b105edb
=> => exporting attestation manifest sha256:26fa5ce51d983e34da5adb46bf5f8597a27da8cfaaaf454952f52ad59d515581
=> => exporting manifest list sha256:bdd082b79605251179a8c5678833fa24adb032d69117a24a74dceec6dbca22db2
=> => naming to docker.io/ramanachandrakiranreddy/k8s-webapp:v2
=> => unpacking to docker.io/ramanachandrakiranreddy/k8s-webapp:v2
```

Docker Push Commands

docker push ramanachandrakiranreddy/k8s-webapp:v1

```
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\V1> docker push ramanachandrakiranreddy/k8s-webapp:v1
The push refers to repository [docker.io/ramanachandrakiranreddy/k8s-webapp]
612c0c1df4c5: Pushed
6a0ac1617861: Pushed
4a8b0b2a5b19: Pushed
781ff90d2644: Pushed
583599bb7d38: Pushed
d2992f529e84: Pushed
82736a35d0e7: Pushed
9d51370795ca: Pushed
aee4e54b3865: Pushed
453da7dbc73e: Pushed
v1: digest: sha256:a8d94fe468a3805e58cb20c24e10b299c471da5cd5152f4b480d40a9884eda38 size: 856
```

docker push ramanachandrakiranreddy/k8s-webapp:v2

```
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\V2> docker push ramanachandrakiranreddy/k8s-webapp:v2
The push refers to repository [docker.io/ramanachandrakiranreddy/k8s-webapp]
781ff90d2644: Layer already exists
4a8b0b2a5b19: Layer already exists
788000cd38da: Pushed
f1001793e05a: Pushed
583599bb7d38: Layer already exists
6a0ac1617861: Layer already exists
82736a35d0e7: Layer already exists
453da7dbc73e: Layer already exists
aee4e54b3865: Layer already exists
612c0c1df4c5: Layer already exists
v2: digest: sha256:bdd082b79605251179a8c5678833fa24adb032d69117a24a74dceec6dbca22db2 size: 856
```

Kubernetes Files

pod.yml

- Creates a single pod
- Runs container using v1 image

```
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl apply -f .\pod.yml
pod/frontend-pod created
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get po
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           12s
```

ReplicaSet (rs.yml)

- Maintains multiple pod replicas
- Ensures high availability

```
PS C:\Users\npci\Downloads\IITH-AI-ML-ASSIGNMENT-main\IITH-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl apply -f .\rs.yml
replicaset.apps/frontend-rs created
PS C:\Users\npci\Downloads\IITH-AI-ML-ASSIGNMENT-main\IITH-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get rs
NAME           DESIRED   CURRENT   READY   AGE
frontend-rs    3         3         3       6s
PS C:\Users\npci\Downloads\IITH-AI-ML-ASSIGNMENT-main\IITH-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get po
NAME           READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           40s
frontend-rs-8db9n  1/1     Running   0           13s
frontend-rs-v6dvd  1/1     Running   0           13s
```

deployment.yml

- Manages application deployment
- Supports rolling updates
- Includes probes

```
PS C:\Users\npci\Downloads\IITH-AI-ML-ASSIGNMENT-main\IITH-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl apply -f .\deployment.yml
deployment.apps/frontend-deployment created
PS C:\Users\npci\Downloads\IITH-AI-ML-ASSIGNMENT-main\IITH-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get deploy
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
frontend-deployment  1/3     3             1           7s
PS C:\Users\npci\Downloads\IITH-AI-ML-ASSIGNMENT-main\IITH-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get deploy
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
frontend-deployment  1/3     3             1           9s
PS C:\Users\npci\Downloads\IITH-AI-ML-ASSIGNMENT-main\IITH-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get deploy
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
frontend-deployment  3/3     3             3           11s
PS C:\Users\npci\Downloads\IITH-AI-ML-ASSIGNMENT-main\IITH-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get po
NAME           READY   STATUS    RESTARTS   AGE
frontend-deployment-db7946498-4qfkh  1/1     Running   0           14s
frontend-deployment-db7946498-8qv2q  1/1     Running   0           14s
frontend-deployment-db7946498-f2qh4  1/1     Running   0           14s
frontend-pod   1/1     Running   0           62s
frontend-rs-8db9n  1/1     Running   0           35s
frontend-rs-v6dvd  1/1     Running   0           35s
```

service.yml

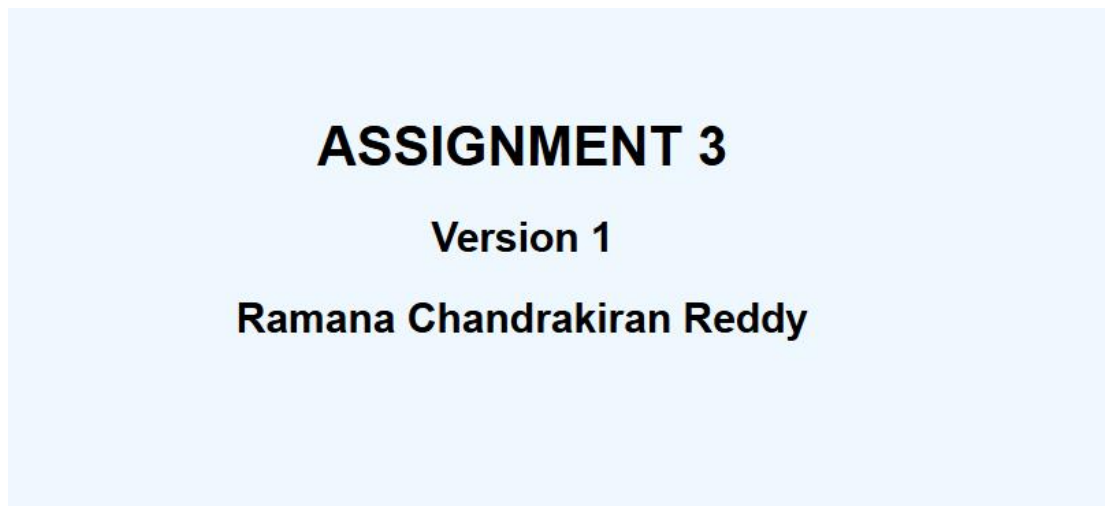
- Exposes application externally
- Routes traffic to pods

```
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl apply -f .\service.yaml
service/frontend-service created
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
frontend-service    NodePort    10.96.89.152   <none>         80:30080/TCP     6s
kubernetes           ClusterIP   10.96.0.1     <none>         443/TCP          29d
```

```
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl port-forward svc/frontend-service 8000:80
Forwarding from 127.0.0.1:8000 -> 80
Forwarding from [::1]:8000 -> 80
```

V1 Running

Browser showing:



Update Deployment to v2

Run image update command:

```
kubectl set image deployment/frontend-deployment
frontend=ramanachandrakiranreddy/k8s-webapp:v2
```

This command updates deployment image from:

- v1 → v2

Monitor Rolling Update

Watch update process:

kubectl get pods -w

During update:

- Old Pods terminate gradually
- New Pods create gradually

Example process:

Old Pod → Terminating

New Pod → Creating

New Pod → Running

```
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl set image deploy/frontend-deployment frontend-ramanachandrakiranreddy/k8s-webapp:v2
deployment.apps/frontend-deployment image updated
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get po
NAME                                READY   STATUS             RESTARTS   AGE
frontend-deployment-84f447d65f-6mtz7 0/1     ContainerCreating   0           0s
frontend-deployment-84f447d65f-tk65q 1/1     Running             0           6s
frontend-deployment-84f447d65f-zvq55 1/1     Running             0           6s
frontend-deployment-db7946d98-4qf4h 0/1     Completed           0           3m
frontend-deployment-db7946d98-f2qhd 0/1     Completed           0           3m
frontend-pod                          1/1     Running             0           3m48s
frontend-rs-8db9n                    1/1     Running             0           3m21s
frontend-rs-v6dvd                    1/1     Running             0           3m21s
```

```
PS C:\Users\npci\Downloads\IITM-AI-ML-ASSIGNMENT-main\IITM-AI-ML-ASSIGNMENT-main\Assignment 3\k8s> kubectl get po
NAME                                READY   STATUS             RESTARTS   AGE
frontend-deployment-84f447d65f-6mtz7 1/1     Running             0           18s
frontend-deployment-84f447d65f-tk65q 1/1     Running             0           24s
frontend-deployment-84f447d65f-zvq55 1/1     Running             0           24s
frontend-pod                          1/1     Running             0           4m6s
frontend-rs-8db9n                    1/1     Running             0           3m39s
frontend-rs-v6dvd                    1/1     Running             0           3m39s
```

v2 Running

Browser showing:

ASSIGNMENT 3

Version 2

Ramana Chandrakiran Reddy